

Machine Learning (CS4102)

Full Marks: 30

Time: 2 Hours

Attempt All Questions.

1. Pick the most appropriate option. (6)
 - (i) Which of the following is the main goal of regression analysis?
 - (a) To classify data into categories
 - (b) To cluster similar data points
 - (c) To predict a continuous outcome variable
 - (d) To reduce the dimensionality of data
 - (ii) Which of the following is a non-linear activation function?
 - a) Sigmoid
 - b) ReLU
 - c) Tanh
 - d) All of the above
 - (iii) Predictive modeling in machine learning refers to:
 - a) Describing historical data patterns only
 - b) Estimating future outcomes based on patterns in data
 - c) Reducing dataset size
 - d) Clustering unlabeled data
 - (iv) Which of the following is a supervised learning algorithm?
 - a) K-Means Clustering
 - b) Principal Component Analysis
 - c) Decision Tree
 - d) Apriori Algorithm
 - (v) The main purpose of unsupervised learning is:
 - a) Predicting continuous values
 - b) Predicting categorical labels
 - c) Finding hidden structures and patterns in data
 - d) Optimizing gradient descent
 - (vi) Which algorithm is used for dimensionality reduction in unsupervised learning?
 - a) PCA
 - b) Decision Tree
 - c) Naive Bayes
 - d) Logistic Regression
2.
 - a) Differentiate between linear regression and logistic regression. (2)
 - b) How do you evaluate the performance of a logistic regression model? (3)
 - c) Explain the role of regularization in logistic regression. (3)
3.
 - a) Draw the neural network structures that implement (i) a 2-input AND gate and (ii) a 2-input OR gate. How do these structures differ from the one required to implement a 2-input XOR gate? (3)
 - b) Compare sigmoid, tanh, and ReLU activation functions in terms of range, gradient behavior, and advantages/disadvantages. (3)
 - c) Explain the vanishing gradient problem. How do ReLU and its variant, such as Leaky ReLU, help to overcome it? (2)
4.
 - a) Why is Cross-Entropy loss function preferred over Sum of Square Error (SSE) loss function for training deep neural networks? (3)
 - b) Explain the difference between stochastic gradient descent, batch gradient descent, and mini-batch gradient descent with examples. (3)
 - c) Explain the working principle of batch normalization and its advantages in deep neural networks. (2)

Machine Learning (CS4102)

Full Marks: 50

Time: 3 Hours

Attempt question number 1 and any FOUR from the rest. (10+10×4)

1.
 - a) Differentiate between supervised and unsupervised learning.
 - b) What is underfitting?
 - c) Differentiate between classification and regression.
 - d) What is a confusion matrix?
 - e) What is a hyperparameter?
2.
 - a) Define the terms bias and variance in a model. Explain the bias–variance tradeoff. (3)
 - b) Define L1 regularization (Lasso Regression) and L2 regularization (Ridge Regression). Describe their effects on model weights. (4)
 - c) What is **Early Stopping**, and how does it act as a regularization method? (3)
- ✓ 3.
 - a) Explain the mathematical expressions and graphs of the ReLU, Sigmoid, and Tanh activation functions, and compare them with one another. (6)
 - b) Explain the Softmax activation function in detail, including its mathematical formulation, working principle, and typical applications in neural networks. (4)
4.
 - a) Describe the architecture of a Deep Neural Network (DNN) and explain how forward propagation and backpropagation are used to train it. (6)
 - b) Describe, with an example, how a deep learning model can be used to capture the semantic representations of sentences. (4)
5.
 - a) Explain the architecture of the VGG16 model in detail with a neat diagram. Additionally, calculate the total number of parameters in both the first convolutional layer and the first pooling layer of the network. (6)
 - b) What is transfer learning and how can VGG16 be used for transfer learning? (4)
6.
 - a) Discuss the different types of noise that can be used in a Denoising Autoencoder. Describe how a Denoising Autoencoder learns to reconstruct clean input from noisy data. (2+3)
 - b) Describe the architecture of a Conditional Variational Autoencoder (CVAE) with a neat diagram. What is the KL divergence term in CVAE, and what is its purpose? (3+2)
- ✓ 7.
 - a) Explain the architecture and working principle of a Recurrent Neural Network (RNN), and discuss the challenges encountered by standard RNNs during training. (3+2)
 - b) Explain the architecture of a Gated Recurrent Unit (GRU) with the help of a diagram, and present the mathematical equations governing its operations, interpreting each term in detail. (5)

Indian Institute of Engineering Science and Technology, Shibpur
BTech 7th Semester Mid-Semester Examination, September 2024

Machine Learning (CS4102)

Full Marks: 30

Time: 2 Hours

Attempt All Questions.

(10×3)

1. a) Estimate the Regressor Coefficients of a Simple Linear Regression Model using Least Square Method. (4)
b) What is the K-Nearest Neighbors (KNN) algorithm, and how does it function in both classification and regression tasks? (3)
c) How is the parameter 'K' chosen in the KNN algorithm? What are the implications of choosing a small vs. a large 'K'? (3)
2. a) Consider the dataset given below and compute the following: (3)
[Given that, $\log_2 3 = 1.6$, $\log_2 5 = 2.3$, $\log_2 7 = 2.8$]

feature object	F ₁	F ₂	Class label
O ₁	1	5	Y
O ₂	1	2	Y
O ₃	2	5	N
O ₄	2	4	N
O ₅	3	4	N

- i) Entropy to classify an object
ii) Entropy to classify an object based on F₂
iii) Information gain based on F₂
- b) Use Naïve Bayes classification algorithm to predict the class label of object X = (F₁ = 2, F₂ = 2). Explain under what circumstances the object cannot be predicted, and provide a solution to resolve the problem. (3)
- c) Explain two popular regularization techniques used to prevent overfitting in logistic regression. (4)
3. a) How do the specific parts of biological neurons correspond to their counterparts in artificial neurons? (2)
b) Draw the structure of a neural network that implements an XOR gate and explain how nonlinear mapping is utilized in this implementation. (3)
c) Describe the perceptron learning algorithm for binary classification problems and prove its convergence. (5)

Indian Institute of Engineering Science and Technology, Shibpur
BTech 7th Semester Final-Semester Examination, November 2024

Machine Learning (CS4102)

Full Marks: 50

Time: 3 Hours

Attempt any FOUR Questions.

(12½ × 4)

1. a) Give the matrix form of a Multiple Linear Regression Model and estimate the Coefficients of it using the Least Squares Method. (5½)
b) Explain the Working Principle of a Support Vector Machine. (7)
2. a) Derive the weight update rule for a sample input $X=(X_1, X_2, \dots, X_n)$ at the output layer of a multi-layer feed forward neural network with $(K-1)$ hidden layers using the backpropagation learning algorithm combined with the gradient descent method. In this context, let M_i denote the number of neurons in the i -th layer, where $i=0, 1, 2, \dots, (K-1), K$. Here, the K -th layer is the output layer, and the 0-th layer represents the input layer. Assume that a sigmoid activation function is used at each layer, and the cost function is the sum of squared errors. Please specify any additional assumptions made in this derivation. (7)
b) What is the cross-entropy loss function, and how does it outperform the quadratic loss function in backpropagation learning for multilayer feedforward neural networks? (5½)
3. a) What are vanishing and exploding gradient problems in neural networks? Discuss their effects and outline potential solutions to these problems. (6½)
b) Explain the Rectified Linear Unit (ReLU) activation function, highlighting its key properties and limitations. How does the Leaky ReLU activation function help address these limitations? (6)
4. a) Justify the properties of “sparsity of connections” and “weight sharing” in Convolutional Neural Networks (CNNs). Discuss the advantages of these properties in the context of CNN performance and efficiency. (3+2½)
b) What is meant by batch size in the context of gradient descent? Explain the different batching strategies used in gradient descent, along with their merits and demerits. How is batch size related to an epoch in these batching strategies? (1+4½ + 1½)
5. a) Briefly discuss the usefulness of the Convolutional layer and the Pooling layer in a Convolutional Neural Network (CNN). Given an input image of size $W \times H \times D$, where W , H , and D are the width, height, and depth respectively, and you are applying N kernels of size $F \times F \times D$ for convolution. If the padding

is P and the stride is S , calculate the volume of the resulting convolved feature map. $(4 + 2\frac{1}{2})$

b) Explain how the regularization techniques of Dropout and Batch Normalization help in addressing the overfitting problem in deep neural networks. (6)

6. a) Construct a high-level diagram of the AlexNet Convolutional Neural Network (CNN) architecture and briefly discuss the function and characteristics of each layer. Additionally, calculate the total number of weights and biases in both the first convolutional layer and the first pooling layer of the network. $(4 + 2\frac{1}{2})$

b) Describe the key steps involved in developing a transfer learning model. Furthermore, discuss the advantages and challenges of employing transfer learning in machine learning applications. $(3+3)$

7. a) Explain the structure of a stacked autoencoder with a diagram and emphasize its key benefits. $(4+2)$

b) What is the primary purpose of a variational autoencoder (VAE), and how does it differ from a simple autoencoder? Explain the reparameterization trick and its significance in VAEs. $(1\frac{1}{2} + 2+3)$

8. a) Explain the main differences between CNN and RNN models. Describe the limitations of an RNN model. $(3+2)$

b) Describe the architecture of LSTM and explain how backpropagation through time is used to train LSTM models. $(4+3\frac{1}{2})$

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech (CST 7th semester) Mid-Semester Examination, September, 2023

Machine Learning: (CS 4102)

Full Marks: 30

Time: 2 hours

Answer any three questions

1. a. Derive a gradient descent training algorithm that minimizes the sum of the squared error cost function, for the following hypothesis:

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_1 x_1^2 + \theta_2 x_2 + \theta_2 x_2^2 + \dots + \theta_n x_n + \theta_n x_n^2$$

where $(x_1, x_2, \dots, x_n)$ represents an instance having n features and $\theta_i, 0 \leq i \leq n$ represents the parameters to be learned. Assume that the number of training instances is m . Give your answer in the form $\theta_j =: \theta_j, + \dots$ for $1 \leq j \leq n$.

- b. What is kernel in SVM and why do we use kernels? What is K-fold cross-validation? How does it affect bias and variance? [7+3]

2. (a) State the role of margin in SVM.

(b) Explain the difference between Hard Margin and Soft Margin for SVM.

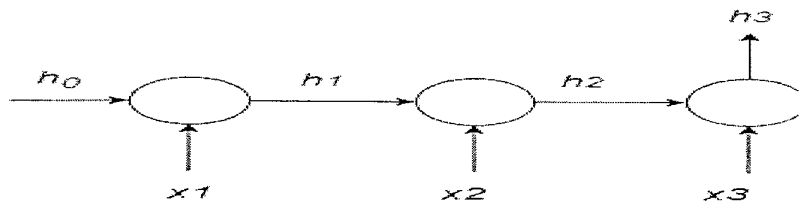
(c) Consider the case of a binary classification starting with training data of 8 tuples, as shown in the following Table. Find parameters W (weight) and b , considering the quadratic programming and KKT constraints to obtain the Lagrange multipliers λ_i for each training tuple. [2+3+5]

x_1	x_2	y	λ
0.38	0.47	+	65.52
0.49	0.61	-	65.52
0.92	0.41	-	0
0.74	0.89	-	0
0.18	0.58	+	0
0.41	0.35	+	0
0.93	0.81	-	0
0.21	0.10	+	0

3. (a) State two reasons why linear regression is not ideal for use in classification. Why mean squared error cost function is not used with logistic regression? Write the cost function that is used instead.

(b) Suppose you are given a model for binary (two-class) logistic regression. Explain briefly (e.g., 2-3 sentences each) how multiclass classification can be performed with the binary classifier using the following strategies: (i.) One vs. All, (ii) One vs. One. If c is the number of classes and m is the number of training examples given, mention how many instances of the binary classifier will be required in each of the above two cases. [5+5]

4. a) Calculate the output y of a three input neuron with bias. The input feature vector is $(x_1, x_2, x_3) = (0.8, 0.6, 0.4)$ and weight values are $[w_1, w_2, w_3, b] = [0.2, 0.1, -0.3, 0.35]$. Use binary sigmoid function as activation function.
- b) Consider a situation where a biased coin with probability of head(p) is tossed N times and the outcomes (head=1, tail=0) are recorded in random variables, x_n , $n=1, 2, \dots, N$. Derive the log likelihood function for estimating parameter p . [3+7]
5. a) Consider a vanilla RNN cell of the form $h_t = \tanh(V \cdot h_{t-1} + W \cdot x_t + b)$. The figure below shows the input sequence x_1, x_2 , and x_3 . Given the dimensions $x_t \in \mathbb{R}^3$ and $h_t \in \mathbb{R}^5$, what is the number of parameters in the RNN cell?



- b) Explain the main difficulties to train RNNs. Specify the ways to handle them. Explain what gating mechanisms are in context of LSTM units.
- c) What is the difference between vanilla RNNs and LSTMs? Why don't we use Long Short Term memory networks for smaller datasets or problems? [3+3+4]

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech (CST 7th Semester) End Semester Examination, November 2023

Machine Learning (CS4102)

Full Marks: 50

Time: 3 Hours

Answer any five questions

1. a) State two reasons why linear regression is not ideal for use in classification. Why is the mean squared error cost function not used with logistic regression? Write the cost function that is used instead.

(b) Suppose you are given a model for binary(two-class) logistic regression. Explain briefly, how multiclass classification can be performed with the binary classifier using the following strategies: (i.) One vs. all, (ii) One vs. One. If c is the number of classes and m is the number of training examples given, mention how many instances of the binary classifier will be required in each of the above two cases.

c) Consider Table 1 showing predictions by a classifier as well as the original labels. There are two classes: positive (p), and negative (n). Calculate the following metrics with respect to class p: i) Precision, ii) Recall, iii) True Positive Rate, and iv) F1 Score. [3+3+4]

No.	Prediction	Actual Class
1	P	P
2	P	n
3	n	n
4	P	n
5	p	P

Table 1: Classifier prediction and actual classes

2) (a) Using the dataset below, build a decision tree that classifies Y as T/F given the binary variables A, B, and C. Draw the tree that the greedy algorithm would learn with zero training error.

A	B	C	Y
F	F	F	F
T	F	T	T
T	T	F	T
T	T	T	F

(b) What is K-fold cross-validation? How does it affect bias and variance?

(c) State two reasons, why linear regression is not ideal for classification? Why mean squared error cost function is not used in logistic regression? Write the cost function that is used instead.

(d) What are the basic differences between feed forward network and a recurrent network. [4+2+2+2]

3) (a) Explain the significance of the RELU activation function in Convolution Neural Network.

(b) Explain Max-Unpooling operation for increasing the resolution of feature maps.

(c) Why do we prefer Convolutional Neural networks (CNN) over Artificial Neural networks (ANN) for image data as input? [4+3+3]

4. a) Calculate the output y of a three-input neuron with bias. The input feature vector is $(x_1, x_2, x_3) = (0.8, 0.6, 0.4)$ and weight values are $[w_1, w_2, w_3, b] = [0.2, 0.1, -0.3, 0.35]$. Use binary Sigmoid function as activation function.

b) Consider a situation where a biased coin with the probability of head(p) is tossed N times and the outcomes (head=1, tail=0) are recorded in random variables, $x_n, n=1, 2, \dots, N$. Derive the log-likelihood function for estimating parameter p . Derive SGD algorithm for solving the max-likelihood problem derived above.

c) There are two common ways to estimate classifier performance i) by using n -fold cross-validation (typically n is 5 or 10) or ii) measure performance on a test set that is distinct from the learning and validation (if used) sets. When will you use cross-validation and when performance on a test set. Provide justification for the answer. [3+3+4]

5. a) What is the vanishing/exploding gradient problem and how is it related to Recurrent Neural Network?

b) Explain Long-term short memory (LSTM) and Gated Recurrent Unit (GRU). How they are different from each other?

c) What is the objective of an Auto Encoder? How auto-encoder is used in dimensionality reduction and image de-noising? What purpose do they serve in comparison to regular auto encoders [3+3+4]

6. a) The Convolutional Neural Network(CNN) has the following database of images below. Each image is of size 100×100 with 3 channels. i.e, the input to the CNN has the dimension of $100 \times 100 \times 3$ and the output has 10 classes. All convolution filters have stride 1 and pooling functions has stride 2. What is the total number of parameters in each of the layer?

Output
Softmax-10
FC1:Fully Connected
P2: 2x2 max pooling
C3:Conv 3x3 -20,pad 2
P1: 2x2 max pooling
C2: Conv 5x5 -20,pad 2
C1: Conv 3x3 -20,pad 1
Input $100 \times 100 \times 3$

Database

b) How do auto encoders handle sparse data? Explain function of auto encoder using sparse data. Are there any specialized auto encoder architectures designed for sparse inputs?

c) i)What is the basic idea behind variational auto encoder?

ii)How have variational auto-encoders been integrated and used in transfer learning? Explain the learning smooth latent state representations of variational auto encoder with example. [3+4+3].

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech (CST 7th semester) Mid-Semester Examination, September 2022

Machine Learning: (CS 4102)

Full Marks: 30

Time: 2 hours

Answer any three questions

1. A bank wants to decide whether a customer can be given a loan, based on two features related to the monthly salary of the customer, and their account balance. For simplicity, we model the two features with two binary variables $X_1, X_2 \in \{0, 1\}$ and the class $Y \in \{0, 1\}$ where $Y = 1$ indicates that the customer can be given a loan, and $Y = 0$ indicates otherwise. Consider the following dataset having four instances:

$(X_1=1, X_2=0, Y=0), (X_1=1, X_2=1, Y=1)$
 $(X_1=0, X_2=0, Y=0), (X_1=0, X_2=1, Y=0)$

- (a) Which model is better for the said application logistic regression or linear regression? Explain briefly.
- (b) Is there any logistic regression classifier using X_1 and X_2 as features, that can perfectly classify the given data?
- (c) If we change the first instance to $(X_1=0, X_2=0, Y=1)$, can there be any logistic regression classifier using X_1 and X_2 as features that perfectly classifies the data? [10]

2. Consider the following data with one input (x) and one output (y): $\{(x=1, y=2), (x=2, y=1), (x=3, y=2)\}$. Apply linear regression to this data, using the hypothesis $h_\theta(x) = \theta_0 + \theta_1 x$, where θ_0 and θ_1 represent the parameters to be learned. Considering learning rate α . Write the iterative steps showing how values of θ_0 and θ_1 are updated in each iteration. Assuming $\alpha = 0.1$, and the initial values $\theta_0 = 1.0, \theta_1 = 0.0$, perform the first three iterations and state the resulting model. Show the steps clearly. [10]

3. (a) State the role of margin in SVM.
- (b) Explain the difference between Hard Margin and Soft Margin for SVM.
- (c) Consider the case of a binary classification starting with training data of 8 tuples, as shown in the following Table.

Using quadratic programming, we can solve the KKT constraints to obtain the Lagrange multipliers λ_i for each training tuple, as shown in the following Table. [2+3+5]

x_1	x_2	y	λ
0.38	0.47	+	65.52
0.49	0.61	-	65.52
0.92	0.41	-	0
0.74	0.89	-	0
0.18	0.58	+	0
0.41	0.35	+	0
0.93	0.81	-	0
0.21	0.10	+	0

4. (a) Using the dataset below, build a decision tree that classifies Y as T/F given the binary variables A, B, and C. Draw the tree that the greedy algorithm would learn with zero training error.

A	B	C	Y
F	F	F	F
T	F	T	T
T	T	F	T
T	T	T	F

- (b) What is K-fold cross-validation? How does it affect bias and variance?
- (c) State two reasons why linear regression is not ideal for classification. Why mean squared error cost function is not used in logistic regression? Write the cost function that is used instead.
- (d) What are the basic difference of feed forward network and recurrent network? [4+2+2+2]

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech (CST) 7th Semester End Semester Examination, November 2022

Machine Learning(CS4102)

Full Marks: 50

Time: 3 Hours

Answer any five questions

1. a) Derive a gradient descent training algorithm that minimizes the sum of squared error cost function, for the following hypothesis:

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_1 x_1^2 + \theta_2 x_2 + \theta_2 x_2^2 + \dots + \theta_n x_n + \theta_n x_n^2$$

where $(x_1, x_2, \dots, x_n)$ represents an instance having n features and $\theta_i, 0 \leq i \leq n$ represent the parameters to be learned. Assume that the number of training instances is m . Give your answer in the form $\theta_j \leftarrow \theta_j + \dots$ for $1 \leq j \leq n$.

- b) Write a stochastic gradient descent algorithm, for optimizing a function $L(\theta) = \frac{1}{N} \sum_{n=1}^N l(x_n, y_n, \theta)$.

- c) Define the expected loss function, clearly mention the random variables. Write the formula for convergence in expectation, clearly define each variable. [4+3+3]

2. a) Consider a support vector machine whose input space is $\mathbb{R}^2$ in which the kernel function is computed as, $k(x, y) = (x \cdot y + 1)^2 - 1$. Let $X_1 = (1, -1, -1)$, $y_1 = -1$, $X_2 = (-3, 1, 1)$, $y_2 = 1$, $X_3 = (-3, 1, -1)$, $y_3 = -1$, $X_4 = (1, 2, 1)$, $y_4 = -1$, and $X_5 = (-1, -1, 2)$, $y_5 = 1$, be five binary labeled training examples. These points are linearly separable. Derive the optimum margin classifier (support vectors, weights and threshold value) and the margin.

- b) What is kernel in SVM and why do we use kernels? [8+2]

3. a) Calculate the output y of a three input neuron with bias. The input feature vector is $(x_1, x_2, x_3) = (0.8, 0.6, 0.4)$ and weight values are $[w_1, w_2, w_3, b] = [0.2, 0.1, -0.3, 0.35]$. Use binary Sigmoid function as activation function.

- b) Consider a situation where a biased coin with probability of head(p) is tossed N times and the outcomes (head=1, tail=0) are recorded in random variables, $x_n, n=1, 2, \dots, N$. Derive the log likelihood function for estimating parameter p . Derive SGD algorithm for solving the max-likelihood problem derived above. [3+7]

4. a) There are two common ways to estimate classifier performance i) by using n -fold cross validation (typically n is 5 or 10) or ii) measure performance on a test set that is distinct from the learning and validation (if used) sets. When will you use cross validation and when performance on a test set. Provide justification for the answer.

- b) What is the objective of auto-encoder?

- c) How auto-encoder is used in dimensionality reduction and image de-noising?

[4+2+4]

5. a) Explain the different layers of Convolutional Neural Network.
b) Explain the significance of ReLU Activation function in CNN.
c) Explain Max-Unpooling operation for increasing the resolution of feature maps. [4+3+3]
6. a) What does the term λ represent in L1 and L2 regularization in deep learning?
b) What is vanishing/exploding gradient problem and how is it related to Recurrent Neural Network?
c) Explain Long-term short memory (LSTM) and Gated Recurrent Unit (GRU). How they are different from each other? [3+3+4]

19m19

Indian Institute of Engineering Science & Technology, Shibpur
Department of Computer Science & Technology

B.Tech Dual Degree 7th Semester Examination, 2019

Subject: Machine Learning

Code: CS721/III

Full Marks- 70

Time: 3hrs

Answer question No. 1 and any three from the rest

1. (i) What are the different steps of Machine Learning Process? 11×2
- (ii) Write differences between supervised and unsupervised learning.
- (iii) Choosing of right Machine Learning solution for your problem depends on which factors?
- (iv) Define the Least square error function, used in ANN for multiclass classification.
- (v) How non-linearly separable problem is converted to linearly separable problem?
- (vi) Define **good margin** in connection with a linear SVM classifier.
- (vii) Calculate number of weights in a fully connected layer of an ANN when images of size, $200 \times 200 \times 3$ are fed as inputs.
- (viii) Write Hebbian learning rule.
- (ix) What are the challenges of Back propagation algorithm?
- (x) When RNN works better than LSTM network?
- (xi) Write the cross entropy cost function.
2. Describe the role of convolution layer, pooling layer and fully connected layer in a CNN. Which activation functions are suitable for the convolution layers and why? 3×3+7
3. What do you mean by Gradient descent and why it is used in machine learning algorithms? What is learning rate and how it is related with minimization of loss function? What do you mean by Stochastic Gradient Descent Algorithm? 3+3+3+4+3
4. Describe the concept of cell state and functions of the gates used in the LSTM network to handle long term dependency of sequence words. What do you mean by back propagation through time? 4+8+4
5. What is the unique feature of Radial Basis Function Neural Network and how it is used for extracting patterns from the dataset? Describe how the Self organizing feature mapping network converts patterns of arbitrary dimensionality into the responses of one or two dimensional arrays of neurons. 2+6+8

3/12/18

Indian Institute of Engineering Science & Technology, Shibpur
Department of Computer Science & Technology

B.Tech Dual Degree 7th Semester Examination, 2018

Subject: Machine Learning

Code: CS723/3

Full Marks- 70

Time: 3hrs

Answer any five questions

All questions carry equal mark

1. (a) What are the implications and benefits of applying convolution operation in the subregions of the input image?
(b) What are the different types of layers used in the convolution neural network (CNN) and explain the purpose and function of using such layers in the CNN architecture? 6+8
2. (a) Why Boltzman machine is called a stochastic, generative model?
(b) How does Restricted Boltzman machine learn to reconstruct the input? 4+10
3. (a) How Vanishing Gradient problem could be tackled in ResNet?
(b) Show the relation between activation size, filter size and depth of each layer in CNN assuming two convolution layers, two pooling layers with max pooling and two fully connected layers. Say, input image size ($m \times m \times r$) and filter size ($n \times n \times q$) where m , n , r and q are variables whose values you assign according to your convenience. 7+7
4. (a) Explain Back propagation through time (BPTT) algorithm of Recurrent neural network.
(b) What are the drawbacks of Recurrent neural network used for sequence modeling? 10+4
5. (a) How gated recurrent units (GRUs) are able to store and filter information using their update and reset gates?
(b) Does GRU solve long term dependence problem in a long sequence? 6+6+2
6. (a) What improvements are there in Long Short Term Memory (LSTM) network with respect to GRU and why?
(b) Give applications of LSTM network. 6+6+2
7. Write short notes: (any two)
(a) Hopfield Neural network (b) Logistic regression and Softmax regression
(c) Associative memory 7+7

5/12/17

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech 7th Semester Examination 2017

Under 5-year Dual-Degree (B. Tech-M. Tech) Programme

Subject: Machine Learning

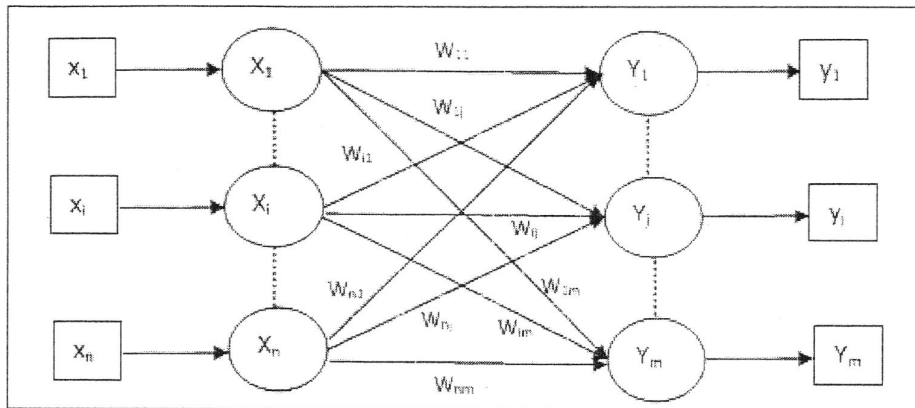
Code: CS723/3

Full marks: 70

Time: 3 hr

Answer any five questions. All questions carry equal marks

1. (a) What is the significance of support vectors in classifying data points using linear support vector machine classifier? If you remove the data points corresponding to the support vectors and non support vectors how the decision boundary changes in each cases?
(b) When describing the placement of decision boundaries using a support vector machine, what function is maximized in Lagrangian formulation of the problem with which constraints?
(c) Suppose using SVM classifier with linear kernel of polynomial degree 2, 100% accuracy is obtained in Training and testing data. Say, you increase the degree of polynomial of this kernel for classifying the same data points. What would you think will happen and why? 6+3+5
2. (a) How states of units are asynchronously updated in Hopfield network? What rule is used in updating the states?
(b) Define the energy function and show that the network converges to a stable state after finite number of iterations. 4+2+3+5
3. (a) What is the main advantage of Radial Basis Function (RBF) network have over a Multi-Layer Perceptron (MLP) network carrying out the same function?
(b) Describe the architecture of a Radial Basis Function network with D input units and K output units. Explain what is computed at each layer with Gaussian function as radial function. 5+3+6
4. (a) What kind of learning is involved in pattern clustering task and why?
(b) Formulate learning rule for condition of a winner node and weight changing rule using the Fig. and symbol given below.
(c) why do we require self-organizing feature map and what is the role of neighboring topology in learning ? 3+5+6



5. (a) Consider a $(32 \times 32 \times 3)$ -image and want to apply a $(5 \times 5 \times 6)$ -filter for convolution. Describe the procedure for generating the activation map. If we repeat the procedure once more using the same filter what will be the outcome. What property of human visual cortex is simulated by this operation?

(b) Explain the role of ReLU activation function, Pooling layer and fully connected layer in a convolution neural network.

6+2+6

6. (a) When does a neural network model become a deep learning model? What are the factors to select the depth of neural network?

(b) What is the problem with sigmoid during back propagation of deep network? How do you solve the problem?

4+4+3+3

7. Write short notes on:(any two)

(a) Reinforcement learning (ii) Associative memory and (iii) Autoencoder

7+7